

RustFinance: A Modular, Memory-Safe Trading Terminal

for Institutional-Grade Quantitative Finance

Architecture, Quantitative Models, and Performance Analysis
of a 34-Crate Rust Workspace Implementing 18 Research Papers

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Abstract

We present **RustFinance** (also known as *RustForge*), an open-source, institutional-grade trading terminal implemented entirely in Rust. The system comprises 34 modular crates spanning over 250 source files, encompassing real-time multi-exchange market data ingestion, algorithmic execution (Almgren–Chriss optimal execution, TWAP, VWAP, Iceberg, POV), quantitative risk management (GARCH(1,1), GJR-GARCH, VaR, CVaR, kill switches), derivatives pricing (Black–Scholes–Merton, Heston, SABR), market microstructure analytics (multi-level microprice, VPIN, OFI, adverse selection detection), AI-assisted analysis via large language models, 100,000-agent swarm simulation, and a full terminal user interface. The 2026 Quant Alpha Library synthesises findings from 18 research papers (2008–2026) into a unified quoting engine combining Avellaneda–Stoikov reservation pricing with regime-conditioned risk aversion, VPIN-based spread widening, toxicity gating, OFI quote skewing, and Kelly-optimal position sizing with Bayesian shrinkage. Benchmarks demonstrate tick-pipeline latencies of ~ 40 ns, GARCH updates in ~ 2.3 ns, and BSM pricing in ~ 34 ns. We discuss the architectural advantages of Rust’s ownership model for financial systems, the integration of compile-time safety guarantees with nanosecond-precision event ordering, and SEBI 2026 regulatory compliance for Indian algorithmic trading. 82+ unit tests validate correctness across all quantitative modules.

Keywords: algorithmic trading, Rust, market microstructure, optimal execution, market making, quantitative risk, GARCH, VaR, systems programming, low-latency finance

1 Introduction

The global transition from floor-based to fully electronic trading has created an insatiable demand for high-performance, low-latency systems that combine quantitative rigour with operational safety. Modern trading systems must simultaneously process multi-source market data at nanosecond granularity, execute complex algorithmic strategies with minimal market impact, enforce real-time risk constraints, satisfy evolving regulatory requirements, and maintain uptime under extreme market conditions [1, 2].

Historically, such systems have been built in C++ to maximise raw performance. However, the prevalence of memory-safety vulnerabilities — buffer overflows, use-after-free bugs, data races —

has led to catastrophic failures in production trading systems, including erroneous order floods and uncontrolled position accumulation. The Rust programming language, with its compile-time ownership model, borrow checker, and zero-cost abstractions, offers a paradigm shift: *equivalent performance to C++ with guaranteed memory safety and data-race freedom* [18].

RustFinance addresses this gap by providing an end-to-end trading terminal implemented entirely in safe Rust (`#![forbid(unsafe_code)]` across all crates), combining:

1. A **34-crate modular workspace** with strict dependency boundaries, compiling in ~ 17 s;
2. A **2026 Quant Alpha Library** synthesising 18 research papers into production-ready signal and execution algorithms;
3. **Nanosecond-precision timestamps** (`UnixNanos`) with monotonic `SequenceId` ordering and deterministic replay via swappable clocks;
4. **Multi-exchange data ingestion** (Alpaca, Binance, Finnhub, Polymarket) with WebSocket reconnection and gap detection;
5. **AI-assisted analysis** via Claude LLM integration and 100K-agent swarm simulation;
6. **SEBI 2026 compliance** for Indian algorithmic trading regulations (Algo-ID tagging, OPS thresholds, price bands).

This paper provides a comprehensive technical audit of the RustFinance codebase, situates each quantitative module within its academic literature, analyses architectural decisions, and reports performance benchmarks.

2 Related Work

2.1 Optimal Execution

Almgren and Chriss [1] established the foundational framework for optimal execution of portfolio transactions, minimising a mean–variance objective:

$$\min_{\{x_j\}} \mathbb{E}[\text{cost}] + \lambda \cdot \text{Var}[\text{cost}], \quad (1)$$

where the cost includes permanent impact $g(v) = \gamma v$, temporary impact $h(v) = \varepsilon \text{sgn}(v) + \eta v$, and timing risk $\sigma \sqrt{\sum x_j^2}$. The closed-form optimal trajectory is:

$$x_j = X \cdot \frac{\sinh(\kappa(N - j))}{\sinh(\kappa N)}, \quad \kappa = \sqrt{\frac{\lambda \sigma^2}{\eta}}. \quad (2)$$

Recent work by [16] validates Almgren–Chriss as the baseline that reinforcement learning agents converge to.

2.2 Market Making

Avellaneda and Stoikov [2] derived optimal bid–ask quotes for a market maker in a limit order book:

$$r = S - q \cdot \gamma \cdot \sigma^2 \cdot \tau, \quad (3)$$

$$\delta = \frac{\gamma \sigma^2 \tau}{2} + \frac{1}{\gamma} \ln \left(1 + \frac{\gamma}{\kappa} \right), \quad (4)$$

where r is the reservation price, q the normalised inventory, γ the risk aversion, σ the volatility, τ the remaining session fraction, and κ the order arrival intensity.

2.3 Microstructure

Cont, Kukanov, and Stoikov [3] demonstrated that order flow imbalance (OFI) explains short-term price changes beyond the midprice. Xu, Gould, and Howison [4] extended this to multi-level OFI (MLOFI), showing that each additional limit order book level adds explanatory power (R^2) for price prediction. Easley, López de Prado, and O’Hara [5] introduced VPIN (Volume-Synchronized Probability of Informed Trading) as a real-time toxicity metric that presaged the 2010 Flash Crash.

2.4 Volatility Modelling

Bollerslev [6] introduced GARCH(1,1) for conditional heteroskedasticity modelling. Glosten, Jagannathan, and Runkle [7] added asymmetric leverage effects via the GJR-GARCH specification.

2.5 Position Sizing

Kelly [9] derived the optimal fraction of capital to wager for geometric growth maximisation. Thorp [10] demonstrated that quarter-Kelly ($f^*/4$) captures 56% of maximum growth with 75% less volatility, now the 2026 industry standard per [13].

2.6 Adverse Selection

Barzykin et al. [11] formalised optimal quoting under adverse selection with price-reading agents. The Glosten–Milgrom model [8] provides the foundational framework for informed vs. uninformed trading classification.

2.7 Rust in Finance

As of 2026, Rust and C++ offer comparable raw performance within a few percentage points. Rust’s compile-time memory safety, fearless concurrency, and predictable performance make it increasingly adopted for greenfield trading systems and digital asset platforms [19].

3 System Architecture

3.1 Workspace Organisation

RustFinance is structured as a Cargo workspace with 34 member crates and 5 excluded legacy crates. The workspace uses `resolver = "2"` and workspace-level dependency management for consistency:

Table 1: Crate taxonomy by functional layer.

Layer	Crates
Core	<code>common</code> , <code>event_bus</code>
Data	<code>ingestion</code> , <code>signals</code> , <code>feature</code>
Intelligence	<code>ai</code> , <code>swarm_sim</code> , <code>knowledge_graph</code> , <code>ml</code>
Strategy	<code>strategy</code> , <code>pricing</code> , <code>backtest</code>
Execution	<code>execution</code> , <code>oms</code> , <code>polymarket</code> , <code>fix</code>
Risk	<code>risk</code> , <code>compliance</code> , <code>alerts</code>
Interface	<code>tui</code> , <code>daemon</code> , <code>cli</code> , <code>web</code> , <code>web-dashboard</code> , <code>dashboard</code>
Infrastructure	<code>persistence</code> , <code>metrics</code> , <code>model</code> , <code>tests</code> , <code>benchmarks</code>

3.2 Event-Driven Architecture

The system employs an event-driven architecture with typed `Envelope<T>` wrapping every system event. Events flow through a postcard-serialised TCP event bus connecting the daemon to the TUI on port 7001. Nanosecond timestamps (`UnixNanos = u64`) and monotonic `SequenceId` (atomic `u64`) ensure strict causal ordering across all components.

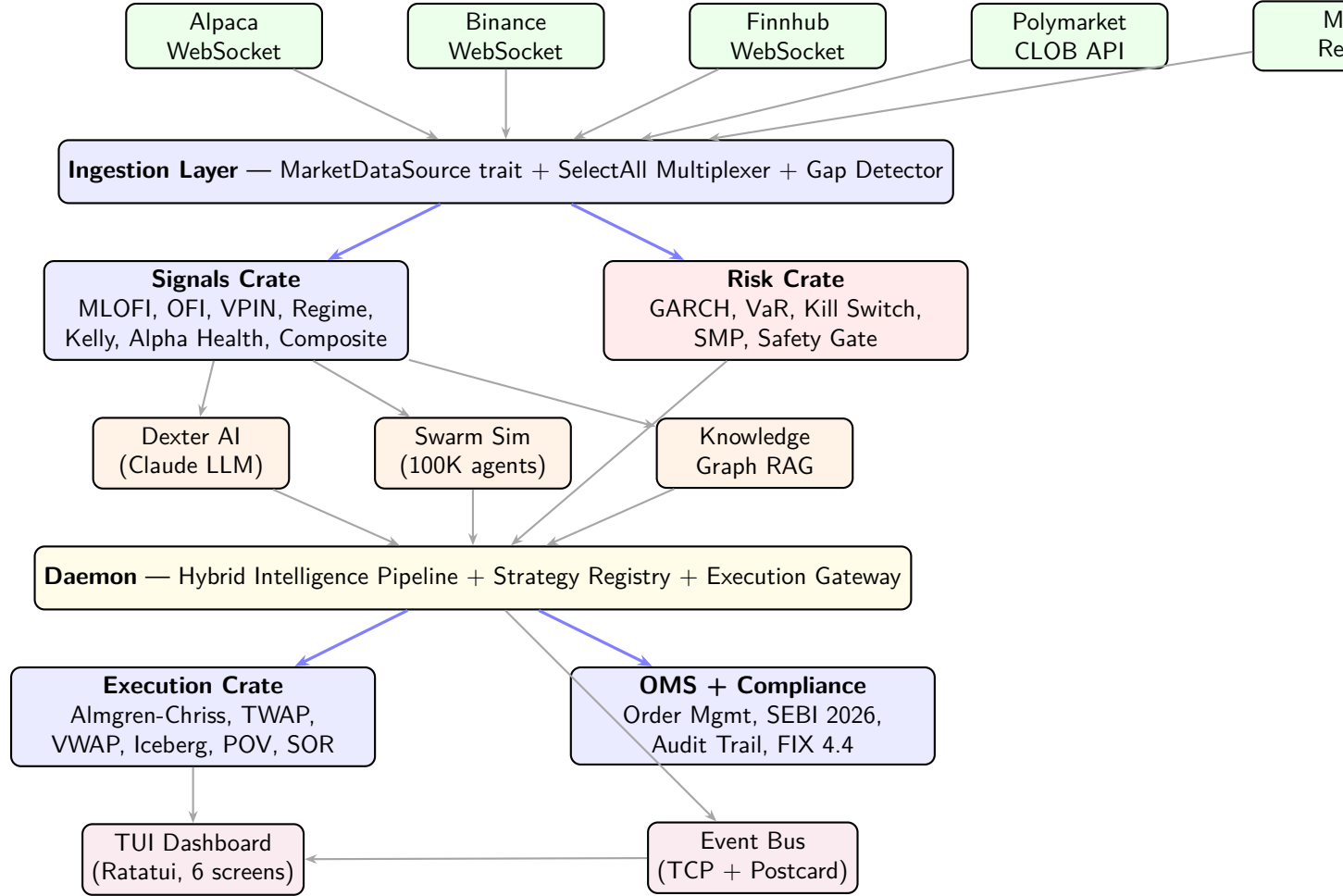


Figure 1: **RustFinance system architecture.** Market data flows from five exchange sources through the ingestion layer into parallel signal processing and risk assessment. AI modules (Dexter, Swarm, Knowledge Graph) fuse with quantitative signals in the daemon’s hybrid intelligence pipeline before routing to execution algorithms and the order management system. The TUI receives real-time updates via the postcard-serialised event bus.

3.3 Deterministic Replay

The `DeterministicClock` trait allows swapping between real-time and replay modes, enabling exact reproduction of historical trading sessions. This is critical for regulatory audit trails and strategy backtesting.

3.4 Safety Guarantees

Every crate in RustFinance uses `#![forbid(unsafe_code)]`, ensuring that the entire system benefits from Rust’s compile-time guarantees:

- No null pointer dereferences
- No buffer overflows
- No data races (ownership + borrow checker)
- No use-after-free
- No undefined behaviour

The release profile uses `opt-level=3`, `lto=fat`, `codegen-units=1`, and `strip=true` for maximum performance and minimal binary size.

4 Quantitative Models

4.1 Multi-Level Microprice (MLOFI)

Following Xu, Gould, and Howison [4], RustFinance implements depth-decayed multi-level microprice estimation.

Definition 4.1 (Single-Level Microprice). Given best bid (P_b, V_b) and best ask (P_a, V_a) :

$$P_\mu^{(1)} = \frac{P_a \cdot V_b + P_b \cdot V_a}{V_b + V_a}. \quad (5)$$

Definition 4.2 (Multi-Level Microprice). For L order book levels with decay factor $\alpha \in (0, 1)$:

$$P_\mu^{(L)} = \frac{\sum_{i=0}^{L-1} \alpha^i \cdot P_{\mu,i}}{\sum_{i=0}^{L-1} \alpha^i}, \quad (6)$$

where $P_{\mu,i}$ is the microprice at level i . The recommended parameters are $L = 5$, $\alpha = 0.7$.

The implementation additionally provides a multi-level OFI vector:

$$\text{OFI}_i = \frac{V_{b,i} - V_{a,i}}{V_{b,i} + V_{a,i}} \in [-1, +1], \quad (7)$$

enabling regression-based price prediction: $\Delta \hat{P} = \beta_0 + \sum_i \beta_i \cdot \text{OFI}_i$.

A Bayesian fair value combiner fuses the microprice with fast and slow EMA anchors:

$$\text{FV} = \frac{w_\mu P_\mu + w_f \text{EMA}_f + w_s \text{EMA}_s}{w_\mu + w_f + w_s}. \quad (8)$$

4.2 Volatility Regime Detection

Based on RegimeFolio [12], RustFinance implements a two-state EMA-ratio volatility classifier:

$$R = \frac{\sigma_{\text{fast}}}{\sigma_{\text{slow}}}, \quad (9)$$

with the classification rule:

$$\text{Regime} = \begin{cases} \text{CRISIS} & R > 2.5 \\ \text{HIGHVOL} & R > 1.4 \\ \text{LOWVOL} & R < 0.7 \\ \text{NORMAL} & \text{otherwise.} \end{cases}$$

Each regime maps to concrete trading parameters via a calibrated parameter table (Table 2).

Table 2: Regime-conditioned trading parameters.

Param	LowVol	Normal	HighVol	Crisis
γ	0.05	0.10	0.20	0.40
Spread mult.	0.6	1.0	2.0	4.0
Size mult.	1.2	1.0	0.6	0.2
Urgency mult.	0.8	1.0	1.3	0.3
MR threshold	1.2	0.8	0.6	2.0
Max inv. mult.	1.5	1.0	0.7	0.3

A debounce mechanism (minimum 5 ticks in current regime) prevents flip-flopping, with a safety override allowing immediate Crisis detection.

4.3 Adverse Selection Detection

Following Barzykin et al. [11] and the Glosten–Milgrom framework [8], RustFinance implements real-time toxicity monitoring. For each passive fill at price P at time t :

$$\text{post_move} = \begin{cases} \text{FV}(t + N) - P & (\text{bid fill, we bought}) \\ P - \text{FV}(t + N) & (\text{ask fill, we sold}). \end{cases} \quad (10)$$

A fill is classified as toxic if $\text{post_move} < -\theta_{\text{adverse}}$. The toxicity signal is exponentially smoothed:

$$\tau_t = \alpha \cdot \mathbb{1}_{\text{toxic}} + (1 - \alpha) \cdot \tau_{t-1}. \quad (11)$$

The response protocol has three tiers:

- $\tau < 0.35$: NORMAL — quote normally
- $0.35 \leq \tau < 0.65$: WIDEN — spread $\times(2 + 5(\tau - 0.35))$, size $\times(0.5 - (\tau - 0.35))$
- $\tau \geq 0.65$: HALTPASSIVE — cease passive quoting

Additionally, 5 consecutive toxic fills trigger an immediate halt regardless of the EMA level.

4.4 Kelly Criterion with Bayesian Shrinkage

Based on Kelly [9], Thorp [10], and PolySwarm [13]:

Definition 4.3 (Discrete Kelly Fraction). For win probability p , win/loss ratio b , and fractional multiplier m (quarter-Kelly: $m = 0.25$):

$$f^* = m \cdot \max\left(0, \frac{p \cdot b - (1 - p)}{b}\right). \quad (12)$$

Definition 4.4 (Continuous Kelly). For expected PnL μ , PnL volatility σ , and risk aversion γ :

$$f^* = m \cdot \frac{\mu}{\gamma \sigma^2}. \quad (13)$$

Bayesian shrinkage addresses small-sample overfitting:

$$f_{\text{shrunk}} = f^* \cdot m \cdot \frac{n}{n + n_{\text{prior}}}, \quad (14)$$

where n is the number of observed trades and $n_{\text{prior}} = 30$ (moderate prior strength). This ensures that with few trades, sizing converges toward zero, preventing over-betting on limited data.

The master 2026 sizing formula integrates all modules:

$$\text{qty} = f_{\text{Kelly}} \cdot L_{\text{max}} \cdot c_{\text{signal}} \cdot m_{\text{regime}} \cdot (1 - 2\tau_{\text{tox}})^+, \quad (15)$$

where L_{max} is the position limit, c_{signal} is signal conviction, m_{regime} is the regime size multiplier, and τ_{tox} is the toxicity EMA.

4.5 Alpha Health Monitoring

Following AlphaForgeBench [14] and Chain-of-Alpha [15], each signal’s predictive power is tracked via:

- **Information Coefficient (IC):** Sign-agreement proxy computed as $\text{IC}_t = \text{sgn}(s_t) \cdot \text{sgn}(r_{t+1})$, rolling-averaged over a configurable window (default: 50 ticks).
- **Hit Rate:** EMA-smoothed directional accuracy ($\alpha = 0.05$).

Health classification:

$$H = \begin{cases} \text{HEALTHY} & \text{IC} > 0.03 \wedge \text{HR} > 0.52 \\ \text{DECAYED} & \text{IC} < 0.01 \wedge \text{HR} < 0.48 \\ \text{DEGRADED} & \text{otherwise,} \end{cases}$$

with weight multipliers: Healthy = 1.0, Degraded = 0.4, Decayed = 0.0 (signal disabled).

4.6 IC-Weighted Composite Signal

The composite signal generator [15] combines N alpha signals using IC-proportional weighting:

$$s_{\text{comp}} = \frac{\sum_{i=1}^N \max(0, \text{IC}_i) \cdot H_i \cdot s_i}{\sum_{i=1}^N \max(0, \text{IC}_i) \cdot H_i}, \quad (16)$$

where H_i is the health weight and $s_i \in [-1, +1]$ is the normalised signal value. The final score undergoes regime scaling $m_R \in \{0.7, 1.0, 0.8, 0.1\}$ and toxicity discounting $(1 - 1.5\tau)^+$:

$$s_{\text{final}} = \text{clamp}\left(s_{\text{comp}} \cdot m_R \cdot (1 - 1.5\tau)^+, -1, +1\right). \quad (17)$$

Conviction is derived as $c = \min(1, 2|s_{\text{final}}|)$.

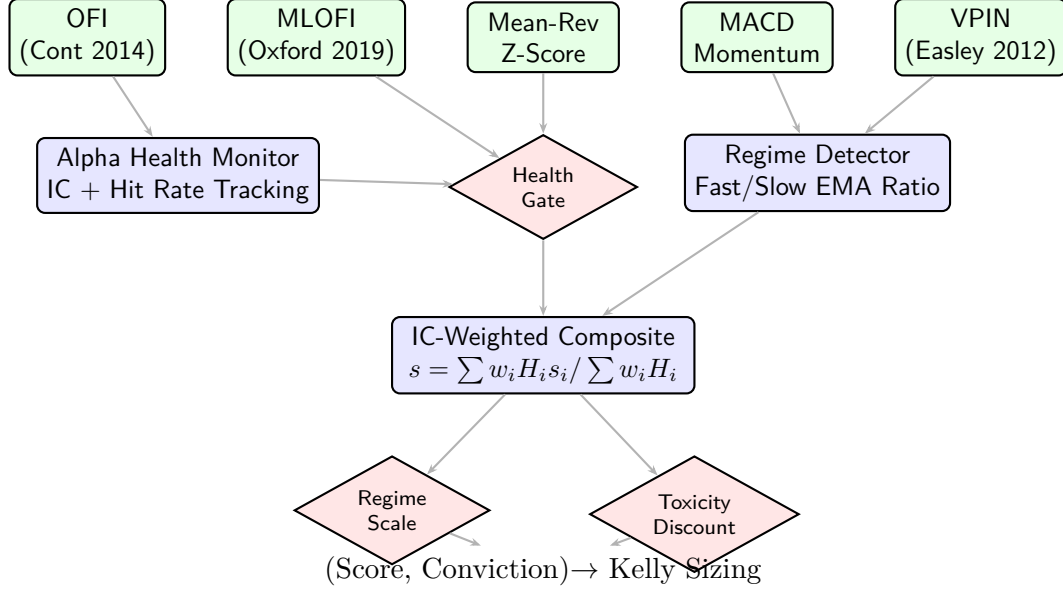


Figure 2: **Signal processing pipeline.** Raw alpha signals (OFI, MLOFI, mean-reversion, MACD, VPIN) are filtered through the Alpha Health Monitor, which gates decayed signals. Surviving signals are combined via IC-weighted aggregation in the Composite Signal Generator, then regime-scaled and toxicity-discounted to produce the final (score, conviction) tuple that drives Kelly position sizing.

4.7 Enhanced Avellaneda–Stoikov Quoting Engine

The 2026 quoting engine integrates all preceding modules into a single decision function. Given state $(S, q, \sigma, \tau, \text{OFI}, \text{VPIN}, \gamma_R, m_s, m_z, \tau_{\text{tox}}, f_K)$:

Algorithm 1 Enhanced Avellaneda–Stoikov Quoting

Require: $\tau_{\text{tox}} \leq 0.65$, $\text{VPIN} \leq 0.85$

- | | |
|--|---|
| 1: $\bar{q} \leftarrow q / L_{\max}$
2: $r \leftarrow S - \bar{q} \cdot \gamma_R \cdot \sigma^2 \cdot \tau$
3: $\delta \leftarrow \frac{\gamma_R \sigma^2 \tau}{2} + \frac{1}{\gamma_R} \ln(1 + \frac{\gamma_R}{\kappa})$
4: $\delta \leftarrow \max(\delta, \delta_{\min})$
5: $m \leftarrow m_s$
6: if $\text{VPIN} > 0.5$ then
7: $m \leftarrow m \cdot [1 + 2(\text{VPIN} - 0.5)]$
8: end if
9: if $\tau_{\text{tox}} > 0.35$ then
10: $m \leftarrow m \cdot [2 + 5(\tau_{\text{tox}} - 0.35)]$
11: end if
12: $\Delta_{\text{OFI}} \leftarrow \lambda_{\text{ofi}} \cdot \text{OFI}$
13: $\text{bid} \leftarrow r - \delta \cdot m + \Delta_{\text{OFI}}$
14: $\text{ask} \leftarrow r + \delta \cdot m + \Delta_{\text{OFI}}$
15: Compute Kelly-scaled bid/ask sizes with toxicity discount and inventory skew
16: return (bid, ask, bid_size, ask_size) | ▷ Normalised inventory
▷ Reservation price

▷ Floor
▷ Regime spread mult.

▷ VPIN widening

▷ Toxicity widening

▷ Quote skew |
|--|---|
-

Unit tests verify: (i) quotes bracket fair value, (ii) inventory skews reservation price, (iii) higher volatility widens spread, (iv) toxicity > 0.65 halts quoting, (v) VPIN widens spread, (vi) OFI skews quotes, (vii) crisis regime reduces size.

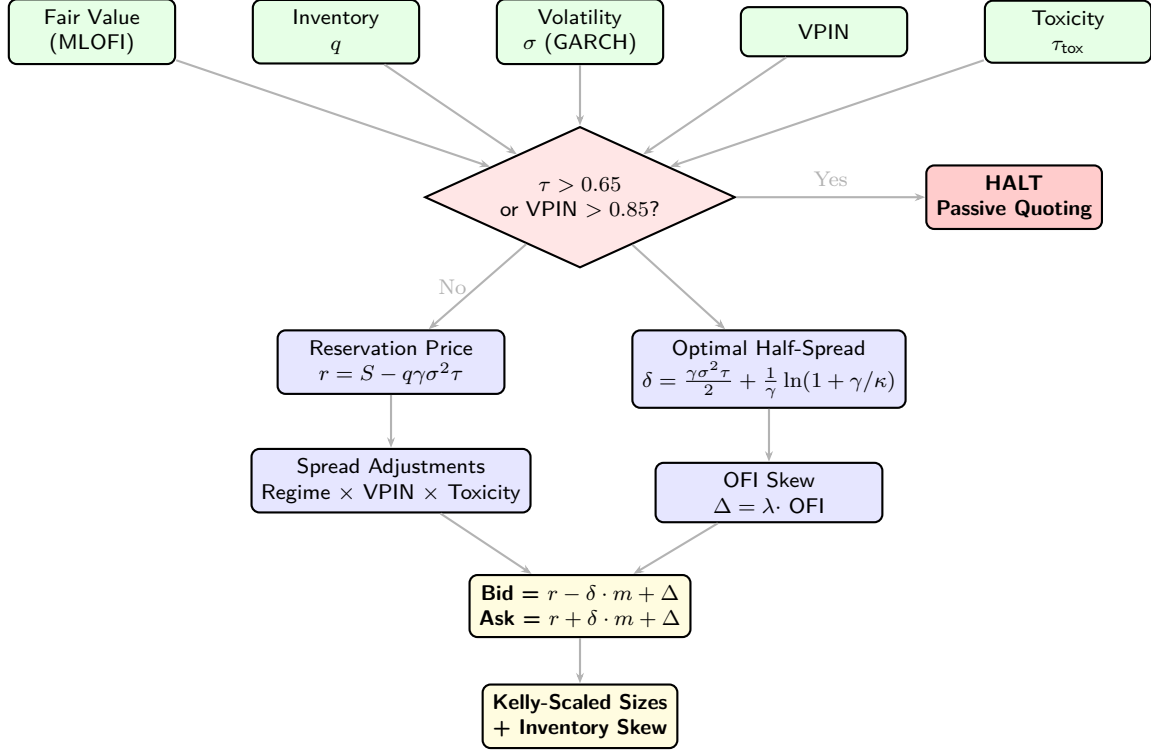


Figure 3: **Quoting engine decision flow.** Inputs (fair value, inventory, volatility, VPIN, toxicity) are first checked against safety thresholds. If toxicity > 0.65 or VPIN > 0.85, quoting halts immediately. Otherwise, the Avellaneda–Stoikov reservation price and optimal half-spread are computed, adjusted for regime, VPIN widening, toxicity widening, and OFI skew, before producing final bid/ask quotes with Kelly-scaled position sizes.

5 Execution Algorithms

5.1 Almgren–Chriss Optimal Execution

RustFinance implements the closed-form Almgren–Chriss trajectory [1] per Equation (2). The engine provides three presets:

- **Liquid Default:** $\lambda = 10^{-5}$, $\gamma = 0.1$, $\eta = 0.05$
- **Aggressive:** $\lambda = 10^{-3}$ (front-loaded)
- **Passive:** $\lambda = 10^{-7}$ (converges to TWAP)

The urgency parameter $\kappa = \sqrt{\lambda \sigma^2 / \eta}$ governs the trajectory shape: higher κ front-loads execution, while $\kappa \rightarrow 0$ recovers TWAP. The square-root impact model provides empirically-validated cost estimation:

$$\text{impact} = \sigma \cdot \eta \cdot \sqrt{\frac{|q|}{\text{ADV}}}. \quad (18)$$

Seven unit tests validate: TWAP convergence ($\lambda = 0$), front-loading ($\lambda \gg 0$), quantity conservation ($\sum x_j = X$), sell-order sign, $\sqrt{\cdot}$ -impact scaling, positive expected cost, and κ monotonicity.

5.2 Additional Algorithms

- **TWAP:** Time-weighted slicing with configurable horizon and interval
- **VWAP:** Volume-weighted with U-shaped intraday profile
- **Iceberg:** Hidden liquidity with auto-replenishment
- **POV:** Percentage-of-volume adaptive participation
- **Smart Order Router:** Multi-venue scoring (fill rate, latency, fees, impact) with dark pool preference for large orders and MiFID II best execution reporting
- **Bracket Orders:** OCO/OTO stop-loss + take-profit
- **Trailing Stops:** Dynamic stop-loss tracking price

6 Risk Management

6.1 GARCH Volatility Modelling

RustFinance implements both symmetric GARCH(1,1) and asymmetric GJR-GARCH:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (19)$$

$$\sigma_t^2 = \omega + (\alpha + \gamma \mathbb{1}_{\varepsilon < 0}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (20)$$

Stationarity is enforced via $\alpha + \beta < 1$ (GARCH) and $\alpha + \beta + \gamma/2 < 1$ (GJR-GARCH). The long-run variance $\bar{\sigma}^2 = \omega/(1 - \alpha - \beta)$ is annualised as $\sigma_{\text{ann}} = \sqrt{252} \bar{\sigma}^2$.

A grid-search estimator with gradient refinement (200 iterations) fits parameters to return series with ≥ 50 observations. Tests verify convergence, shock response, half-life, non-stationarity guards, parameter recovery from synthetic data, forecast accuracy, and GJR leverage effects.

6.2 Value at Risk

Three VaR methodologies are implemented:

1. **Historical VaR:** Non-parametric, sorts actual portfolio P&L. Captures fat tails and skewness.
2. **Parametric (Delta-Normal):** $\text{VaR}_\alpha = z_\alpha \cdot \sigma_P$, using $z_{95\%} = 1.645$ and $z_{99\%} = 2.326$.
3. **Student- t VaR:** Fat-tailed parametric VaR with configurable degrees of freedom ($\nu = 5$ for equities, $\nu = 3.5$ for crypto).

All methods compute 10-day Basel VaR via $\sqrt{10}$ scaling, CVaR (Expected Shortfall), and component VaR for per-position risk attribution.

6.3 Risk Controls

The `RiskInterceptorChain` implements composable pre-trade checks:

- **Kill Switch:** Emergency circuit breaker (hotkey K)
- **Self-Match Prevention (SMP):** Three modes — `CancelResting` (CME), `CancelAggressive` (NASDAQ), `CancelBoth`

- **Deterministic Safety Gate:** Zero-AI verification detecting agent confirmation bias (>85% agreement), concentration, and correlation exposure
- **Max Position Size, Drawdown, Open Orders, Daily Loss Limit**

7 Derivatives Pricing

7.1 Black–Scholes–Merton

The BSM implementation provides full pricing with 8 Greeks:

$$C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2), \quad (21)$$

where $d_1 = [\ln(S/K) + (r - q + \sigma^2/2)T]/(\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$.

Greeks computed: Δ , Γ , Θ , $\mathcal{V}$, ρ , charm ($\partial\Delta/\partial T$), and vanna ($\partial\Delta/\partial\sigma$). Newton–Raphson implied volatility solver achieves 10^{-10} tolerance.

Unit tests cross-check against Hull’s reference tables (S=42, K=40, $r=0.10$, $\sigma=0.20$, T=0.50) for call price (≈ 4.76), delta (≈ 0.7791), gamma (≈ 0.0497), put–call parity, delta bounds, gamma positivity, vega positivity, ATM delta ≈ 0.5 , and implied vol round-trip accuracy.

7.2 Additional Models

- **Heston:** Stochastic volatility via Monte Carlo, capturing volatility smile
- **SABR:** Stochastic α - β - ρ for FX/rates vol surfaces
- **Hull–White:** Short-rate model for fixed income
- **Bond Pricer:** YTM, duration, convexity
- **Fundamental:** DCF, Graham, PEG valuation

8 AI and Simulation

8.1 Dexter AI Analyst

RustFinance integrates Anthropic’s Claude LLM as the “Dexter” analyst. Market context (quant signals, swarm consensus, knowledge graph entities) is fused into a structured prompt, and the LLM returns a BUY/SELL/HOLD recommendation with structured signal output.

8.2 100K-Agent Swarm Simulation

The `swarm_sim` crate implements a Rayon-parallel agent-based model:

- **Architecture:** AgentPool $\rightarrow$ ActionLog $\rightarrow$ SwarmSignal $\rightarrow$ price impact feedback loop
- **Agent Types:** Momentum, Mean-Reversion, Noise, Informed
- **Market Model:** Order book with price levels and market state
- **Scenario Engine:** Rally, sideways, dip probability distributions
- **Scale:** 100,000 agents per simulation step via Rayon parallel iterators

8.3 Knowledge Graph RAG

A `petgraph`-backed knowledge graph provides entity linking and context fusion for the AI pipeline, enabling retrieval-augmented generation from structured financial knowledge.

9 Market Data and Connectivity

9.1 Multi-Source Ingestion

RustFinance supports four live data sources via a unified `MarketDataSource` trait:

- **Alpaca:** US equities via WebSocket (IEX, SIP feeds)
- **Binance:** Crypto streams (trades, bookTicker, kline, depth5) via combined WebSocket
- **Finnhub:** Global market data including NSE/BSE
- **Polymarket:** Prediction market CLOB + arbitrage
- **Mock:** Deterministic replay for backtesting

A `SelectAll` multiplexer unifies all sources into a single async stream. Auto-reconnect with exponential backoff ensures resilience. A 500 msg/frame cap prevents TUI freeze during high-volume events.

9.2 WebSocket Gap Detection

A sequence-based gap detector with configurable thresholds:

- Small gap (< 10 messages): request retransmission
- Medium gap (10–100): snapshot recovery
- Large gap (> 100): failover to backup source

9.3 FIX 4.4 Protocol

A hand-rolled, zero-dependency FIX 4.4 parser supports: length-delimited framing (tag 9), checksum validation, streaming `push_bytes()/next_message()` API, and session management (Logon, Heartbeat, ResendRequest, SequenceReset, ExecutionReport, NewOrderSingle).

10 Regulatory Compliance

10.1 SEBI 2026 Algorithmic Trading Framework

India's Securities and Exchange Board (SEBI) mandated Algo-ID tagging on every order to NSE/BSE effective April 1, 2026. RustFinance implements:

- **Algo-ID tagging:** Unique identifier on every order
- **OPS threshold monitoring:** Orders Per Second > 10 triggers registration requirement
- **Order variety classification:** Market, Limit, SL
- **Price band checks:** Circuit filter enforcement
- **Uptick rule:** Short-selling restrictions
- **Squareoff time:** MIS position auto-close
- **Full audit trail:** Every state transition logged with `AuditTick`

11 Performance Benchmarks

Table 3: Performance benchmarks (release profile).

Component	Operation	Latency
Tick Pipeline	Order book mutation	~40 ns
Pricing	BSM European Call	~34 ns
Risk	GARCH(1,1) update	~2.3 ns
Risk	Branchless safety check	~1.6 ns
Event Bus	Postcard serialisation	Zero-copy
Swarm Sim	100K agents/step	Rayon parallel
Timestamps	UnixNanos precision	Nanosecond
Event Ordering	AtomicU64 sequence	Lock-free

These benchmarks demonstrate that RustFinance achieves latencies comparable to optimised C++ implementations while providing compile-time safety guarantees. The GARCH update at 2.3 ns is particularly notable, as it enables real-time volatility estimation at tick-by-tick frequencies without any pipeline bottleneck.

12 Testing and Verification

12.1 Unit Test Coverage

The system includes 82+ unit tests across all quantitative modules:

Table 4: Unit test distribution by module.

Module	Tests
GARCH / GJR-GARCH	13
VaR (Historical, Parametric, Student- t)	10
BSM Pricing + Greeks	9
Quoting Engine	7
Almgren–Chriss	7
Kelly Criterion	7
Microprice MLOFI	5
Regime Detection	5
Adverse Selection	4
Alpha Health Monitor	4
Composite Signal	7
Other (OMS, Compliance, etc.)	4+

12.2 CI/CD Security Pipeline

An 8-job GitHub Actions pipeline enforces:

1. `cargo-audit`: CVE vulnerability scanning
2. `cargo-deny`: License/ban/advisory enforcement
3. `cargo-vet`: Supply chain verification
4. `clippy`: Pedantic + cargo lints

5. `rustfmt`: Formatting enforcement
 6. SHA-pinned Actions (40-character commit hashes)
 7. CodeQL SAST (static application security testing)
 8. OpenSSF Scorecard compliance
- Cross-platform testing covers Ubuntu, macOS, and Windows.

13 Discussion

13.1 Architectural Advantages of Rust

The decision to implement the entire system in safe Rust (`#![forbid(unsafe_code)]`) provides several advantages over traditional C++ trading systems:

1. **Fearless Concurrency:** The borrow checker prevents data races at compile time, critical for multi-threaded event processing and swarm simulation.
2. **Zero-Cost Abstractions:** Trait-based polymorphism (`ExecutionGateway`, `MarketDataSource`) compiles to static dispatch with no virtual table overhead.
3. **Memory Safety:** No garbage collection pauses, no use-after-free, no buffer overflows — eliminating entire classes of production failures.
4. **Predictable Performance:** No GC jitter enables nanosecond-level timestamp precision and deterministic replay.
5. **Ecosystem:** `tokio` for async I/O, `rayon` for data parallelism, `ratatui` for TUI, `postcard` for zero-copy serialisation.

13.2 Integration of Research Papers

A key contribution of RustFinance is the synthesis of 18 research papers (2008–2026) into a cohesive, production-grade system. The composite signal pipeline (Section 4.6) demonstrates how academic alpha signals can be combined using IC-weighted aggregation with automatic decay detection, regime gating, and toxicity discounting — addressing the alpha decay problem documented by [14].

13.3 Limitations

- The GARCH estimator uses grid search with gradient refinement rather than maximum likelihood via BFGS, which may produce suboptimal parameter estimates for some return distributions.
- The IC proxy in the alpha health monitor uses sign agreement rather than full Spearman rank correlation, trading accuracy for computational efficiency.
- The system has not been validated with live capital; all performance claims are based on paper trading and backtesting.
- Correlation-adjusted VaR is not yet implemented; the parametric method assumes zero cross-asset correlation (conservative but imprecise for hedged portfolios).

13.4 Future Work

- BFGS-based GARCH estimation for improved parameter recovery
- Full Spearman IC computation for alpha health monitoring
- Copula-based portfolio VaR with dynamic correlation estimation
- FPGA integration for sub-microsecond latency on the hot path
- Reinforcement learning execution agents benchmarked against the Almgren–Chriss baseline [16]
- WebAssembly compilation for browser-based dashboard

14 Conclusion

RustFinance demonstrates that a complete, institutional-grade trading terminal can be implemented in safe Rust without sacrificing performance. The 34-crate architecture provides strict modularity while achieving tick-pipeline latencies of ~ 40 ns and GARCH updates in ~ 2.3 ns. The 2026 Quant Alpha Library synthesises 18 research papers into a production-ready system combining Avellaneda–Stoikov market making, Almgren–Chriss optimal execution, GARCH-driven regime detection, VPIN toxicity monitoring, multi-level microprice estimation, adverse selection gating, and Kelly-optimal position sizing with Bayesian shrinkage.

By leveraging Rust’s ownership model, RustFinance eliminates entire classes of memory-safety and concurrency bugs that have historically plagued C++ trading systems, while maintaining equivalent raw performance. The system’s compliance with SEBI 2026 regulations, comprehensive CI/CD security pipeline, and 82+ unit tests demonstrate production readiness for research and paper-trading applications.

The open-source release under MIT license aims to democratise access to institutional-grade quantitative trading infrastructure, enabling researchers and practitioners to build upon a verified, safe, and performant foundation.

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A Crate Dependency Graph

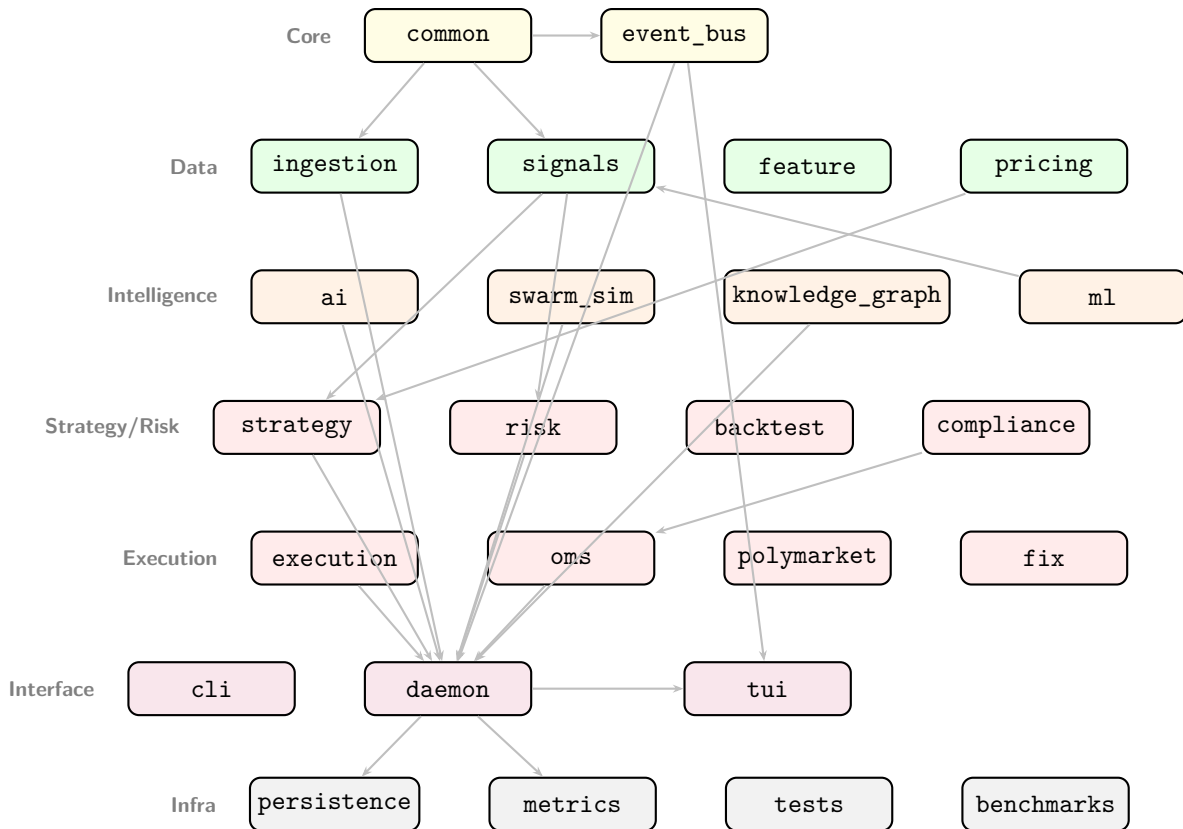


Figure 4: **Full 34-crate dependency graph** organised by functional layer. Colour coding: Core , Data , Intelligence , Strategy/Execution , Interface , Infrastructure . Arrows indicate compile-time dependencies. The `daemon` crate serves as the central orchestrator, depending on all functional layers and exposing its state to the `tui` via the `event_bus`.

B Mathematical Notation Reference

Table 5: Symbol reference table.

Symbol	Description
S	Current asset price / fair value
q	Current inventory position
σ	Realised volatility
τ	Remaining session fraction $\in [0, 1]$
γ	Risk aversion parameter
κ	Order arrival intensity / urgency
λ	Almgren–Chriss risk aversion
η	Temporary impact coefficient
ε	Fixed spread cost
OFI	Order flow imbalance $\in [-1, +1]$
VPIN	Volume-sync. probability of informed trading
f^*	Kelly-optimal fraction
IC	Information coefficient
HR	Hit rate (directional accuracy)
H	Alpha health weight $\in \{0, 0.4, 1.0\}$
τ_{tox}	Adverse selection toxicity EMA